

EE3621: Digital Signal Processing

Lecture 18: IIR Filter — Cascade Form Realization (III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 06:00 (6 mins):** Motivation for Modular Decomposition: Why Direct Forms fail in high-order systems (extreme root sensitivity of high-degree polynomials to coefficient quantization).
 - **06:00 – 16:00 (10 mins):** Mathematical Foundations of Cascade Realization: Factoring $H(z)$ into 1st and 2nd-order sections (SOS / Biquads), pairing real and complex conjugate poles/zeros to ensure real arithmetic coefficients.
 - **16:00 – 25:00 (9 mins):** Canonical Biquad Architecture (Direct Form II SOS): Shared central delay spine, state update and output equations, multiplier/adder budgeting, and intermediate state links.
 - **25:00 – 32:00 (7 mins):** Practical Engineering Guidelines: Optimal Pole-Zero Pairing ($\theta_z \approx \theta_p$ for peak resonance suppression), Section Ordering (low- Q to high- Q), and Dynamic Range Gain Scaling (g_k).
 - **32:00 – 38:00 (6 mins):** Comprehensive Worked Numerical Examples: 4th-order cascade realization (algebraic derivation, difference equations, SFG verification) and odd-order (3rd-order) systems.
 - **38:00 – 40:00 (2 mins):** Faculty Insights, Common Misconceptions, and University Semester Exam Review.
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1 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Formulate** the factorized rational transfer function $H(z) = g \prod_{k=1}^K H_k(z)$ from high-order numerator and denominator polynomials.
2. **Explain** why cascade realization dramatically reduces coefficient quantization sensitivity compared to Direct Form I and Direct Form II (Kaiser's root sensitivity theorem).
3. **Synthesize** canonical second-order Direct Form II sections (biquads) with minimum memory registers ($\max(M_k, N_k) \leq 2$).
4. **Apply** optimal pole-zero pairing criteria to flatten section frequency responses and prevent internal register saturation.

5. **Order** cascaded biquad sections to maximize signal-to-noise ratio (SNR) and prevent dynamic range overflow.
 6. **Audit** total hardware complexity (multipliers, adders, delay registers) for canonical and non-canonical cascade structures.
 7. **Realize** odd-order IIR filters using a modular combination of first-order and second-order sections.
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2 2. MATHEMATICAL FOUNDATIONS OF CASCADE REALIZATION

2.1 2.1 Why Not High-Order Direct Forms? (The Root Sensitivity Catastrophe)

In Lecture 17, we studied the Direct Form I and Direct Form II realizations of general IIR systems:

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} \quad (1)$$

While direct form structures are mathematically sound and straightforward to implement for low orders ($N \leq 2$), they suffer from a **fatal practical limitation in finite-precision digital hardware**: *extreme sensitivity of root locations to coefficient quantization*.

Consider a high-degree polynomial representing the denominator $A(z) = \prod_{i=1}^N (1 - p_i z^{-1})$. When implemented in fixed-point DSP registers (e.g., 16-bit or 24-bit words), each coefficient a_k is quantized to $\hat{a}_k = a_k + \Delta a_k$.

According to **Kaiser's Root Sensitivity Theorem**, the displacement Δp_i of the i -th pole due to an error Δa_k in the k -th coefficient is given by:

$$\frac{\partial p_i}{\partial a_k} = - \frac{p_i^{N-k}}{\prod_{\substack{m=1 \\ m \neq i}}^N (p_i - p_m)} \quad (2)$$

Notice the denominator product:

$$\prod_{m \neq i} (p_i - p_m) \quad (3)$$

When a filter has high selectivity (such as a sharp cutoff low-pass or narrowband band-pass filter), multiple poles p_m cluster tightly together near the unit circle ($|p_i| \approx 1$). Consequently:

- The Euclidean distance between adjacent poles ($p_i - p_m$) becomes exceptionally small ($\ll 1$).
- Their product over high order N shrinks to near zero: $\prod_{m \neq i} (p_i - p_m) \rightarrow 0$.
- The partial derivative $\left| \frac{\partial p_i}{\partial a_k} \right|$ explodes exponentially with order N !

As a devastating result, a minuscule rounding error in the least significant bit of a single coefficient a_k can displace a stable pole inside the unit circle ($|p_i| < 1$) across the boundary into the unstable region ($|p_i| \geq 1$), causing catastrophic unbounded oscillation.

The Cascade Breakthrough: By factoring the N -th order polynomial into a product of independent first- and second-order polynomials, each coefficient a_{k1}, a_{k2} controls **only its own two localized roots**. The denominator product in Kaiser's formula involves at most one other root ($|p_1 - p_2|$), completely decoupling the roots and reducing root sensitivity by orders of magnitude!

2.2 General Cascade Decomposition Formulation

Any rational IIR transfer function with real coefficients can be factored into a product of second-order sections (SOS, commonly called **Biquads**) and at most one first-order section (if the order is odd):

$$H(z) = g \prod_{k=1}^K H_k(z) = g \prod_{k=1}^K \frac{b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}} \quad (4)$$

where:

- $K = \lceil N/2 \rceil$ is the total number of cascaded sections.
- g is an overall scalar gain factor extracted such that $b_{k0} = 1$ (or normalized for dynamic range scaling).
- Each section $H_k(z)$ has a numerator of degree $M_k \leq 2$ and a denominator of degree $N_k \leq 2$.
- For an odd-order filter, one section has $b_{k2} = 0$ and $a_{k2} = 0$, reducing to a first-order section.

2.3 Handling Real vs. Complex Conjugate Roots

Because the physical filter must operate using real arithmetic in hardware (multipliers and adders cannot process imaginary numbers without $4\times$ computational overhead), all coefficients a_{ki} and b_{ki} must be **strictly real**.

2.3.1 Case 1: Complex Conjugate Pole Pairs

Complex poles always occur in conjugate pairs for real-coefficient filters:

$$p_k = r_k e^{j\theta_k}, \quad p_k^* = r_k e^{-j\theta_k} \quad (5)$$

Combining their factors algebraically yields a second-order polynomial with real coefficients:

$$\begin{aligned} D_k(z) &= (1 - p_k z^{-1})(1 - p_k^* z^{-1}) \\ &= 1 - (p_k + p_k^*)z^{-1} + (p_k p_k^*)z^{-2} \\ &= 1 - 2r_k \cos(\theta_k)z^{-1} + r_k^2 z^{-2} \end{aligned} \quad (6)$$

Comparing with $1 + a_{k1}z^{-1} + a_{k2}z^{-2}$:

$$\mathbf{a}_{k1} = -2\mathbf{r}_k \cos(\theta_k), \quad \mathbf{a}_{k2} = \mathbf{r}_k^2 \quad (7)$$

For stability, the pole radius must satisfy $r_k < 1$, which directly imposes the coefficient bounds:

$$0 < a_{k2} < 1, \quad |a_{k1}| < 1 + a_{k2} \quad (8)$$

2.3.2 Case 2: Real Pole Pairs

If two real poles $p_1, p_2 \in \mathbb{R}$ are present, they are grouped together into a single biquad:

$$(1 - p_1 z^{-1})(1 - p_2 z^{-1}) = 1 - (p_1 + p_2)z^{-1} + (p_1 p_2)z^{-2} \quad (9)$$

where $a_{k1} = -(p_1 + p_2)$ and $a_{k2} = p_1 p_2$.

2.3.3 Case 3: Complex Conjugate and Real Zeros

Similarly, zeros $z_k = r_{zk}e^{\pm j\theta_{zk}}$ combine into:

$$1 - 2r_{zk} \cos(\theta_{zk})z^{-1} + r_{zk}^2 z^{-2} \quad (10)$$

yielding real numerator coefficients $b_{k1} = -2r_{zk} \cos(\theta_{zk})$ and $b_{k2} = r_{zk}^2$.

2.4 2.4 Modular Signal Interconnection

In the cascade realization, the output of the $(k-1)$ -th biquad serves directly as the input to the k -th biquad:

$$x[n] \rightarrow \boxed{g} \xrightarrow{x_1[n]} \boxed{H_1(z)} \xrightarrow{y_1[n]=x_2[n]} \boxed{H_2(z)} \xrightarrow{y_2[n]=x_3[n]} \dots \xrightarrow{y_{K-1}[n]=x_K[n]} \boxed{H_K(z)} \rightarrow y[n] \quad (11)$$

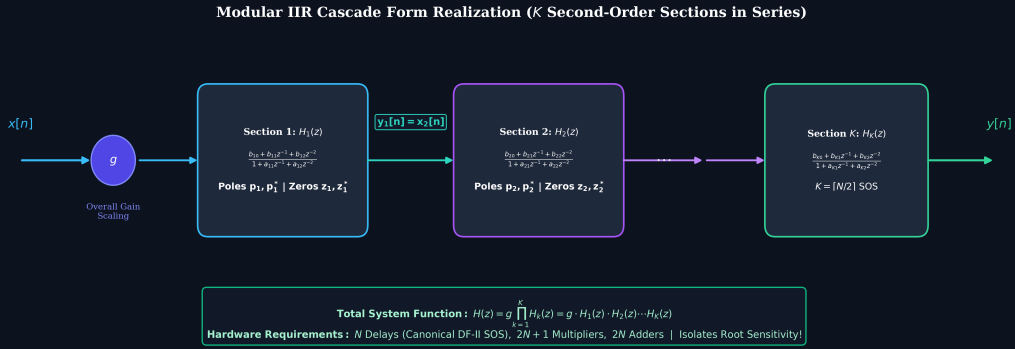


Figure 1: Modular IIR Cascade Architecture. The high-order system is decomposed into a chain of $K = \lceil N/2 \rceil$ second-order sections (biquads) interconnected in series, where each intermediate link satisfies $x_k[n] = y_{k-1}[n]$.

3 3. CANONICAL BIQUAD ARCHITECTURE (DIRECT FORM II SOS)

Each individual second-order section $H_k(z)$ in (4) can be realized using any LTI structure. In modern digital signal processors and VLSI ASICs, the **Direct Form II Canonical Biquad** is universally adopted because it minimizes register count.

3.1 3.1 Signal Flow Graph and Mathematical Derivation

The transfer function of the k -th biquad is:

$$H_k(z) = \frac{Y_k(z)}{X_k(z)} = \frac{b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}} \quad (12)$$

By defining the intermediate state sequence $W_k(z)$ such that:

$$W_k(z) = \frac{X_k(z)}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}} \implies X_k(z) = W_k(z) + a_{k1}z^{-1}W_k(z) + a_{k2}z^{-2}W_k(z) \quad (13)$$

and the output:

$$Y_k(z) = (b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}) W_k(z) \quad (14)$$

Taking the inverse Z -transform yields the two fundamental computational equations of the biquad:

$$w_k[n] = x_k[n] - a_{k1}w_k[n-1] - a_{k2}w_k[n-2] \quad (\text{Biquad State Equation}) \quad (15)$$

$$y_k[n] = b_{k0}w_k[n] + b_{k1}w_k[n-1] + b_{k2}w_k[n-2] \quad (\text{Biquad Output Equation}) \quad (16)$$

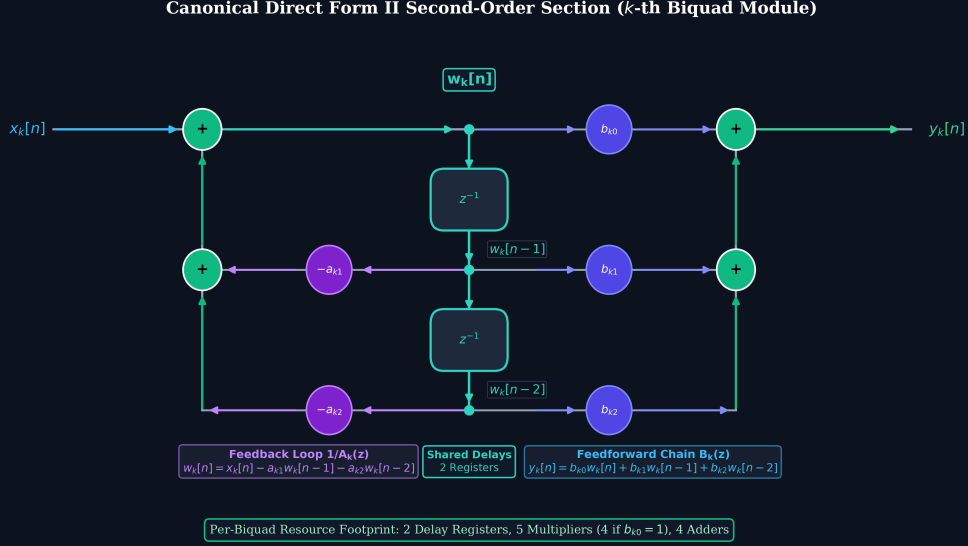


Figure 2: Canonical Direct Form II Second-Order Section (SOS / Biquad). The central delay spine (z^{-1}, z^{-1}) stores the intermediate state variables $w_k[n-1]$ and $w_k[n-2]$, simultaneously serving both the feedback loop (left) and the feedforward chain (right).

3.2 Hardware Resource Audit per Biquad

For each individual canonical second-order section:

- **Delay Registers (z^{-1}):** Exactly **2** registers (storing $w_k[n-1]$ and $w_k[n-2]$).
- **Multipliers:** Up to **5** multipliers ($b_{k0}, b_{k1}, b_{k2}, -a_{k1}, -a_{k2}$). When normalized ($b_{k0} = 1$), this reduces to **4** non-trivial multipliers.
- **Adders:** Exactly **4** two-input adders (2 in the feedback summation, 2 in the feedforward summation).

If a non-canonical Direct Form I biquad were used instead, each section would require $M_k + N_k = 2 + 2 = 4$ delay registers. Over a 6th-order filter ($K = 3$), DF-I cascade requires 12 delay registers, whereas canonical DF-II cascade requires only **6 delay registers**—a massive 50% memory savings.

4. PRACTICAL ENGINEERING GUIDELINES: PAIRING, ORDERING, AND SCALING

While all pairings and orderings yield mathematically identical transfer functions under infinite precision, they exhibit **vastly different performance** under finite-precision fixed-point arithmetic! Three critical engineering rules must be followed:

4.1 4.1 Optimal Pole-Zero Pairing ($\theta_z \approx \theta_p$)

A filter with K sections has $(K!)^2$ possible ways to pair poles and zeros.

- **The Phenomenon:** A pole pair close to the unit circle ($r_k \approx 1$) creates a sharp, resonant peak in the frequency response $|H_k(e^{j\omega})|$. Conversely, a transmission zero creates a deep attenuation notch.
- **The Criterion:** Pair each complex pole pair with the **spatially closest zero pair** in the z -plane ($\theta_z \approx \theta_p$).
- **Engineering Rationale:** The notch from the zero cancels out the sharp resonance peak of the adjacent pole within that specific biquad. This flattens the individual section's frequency response $|H_k(e^{j\omega})|$, minimizing the peak-to-valley dynamic range ratio:

$$\min \left(\frac{\max_{\omega} |H_k(e^{j\omega})|}{\min_{\omega} |H_k(e^{j\omega})|} \right) \quad (17)$$

Preventing high resonance peaks eliminates internal state register overflow, while eliminating deep notches keeps signals well above the quantization noise floor.

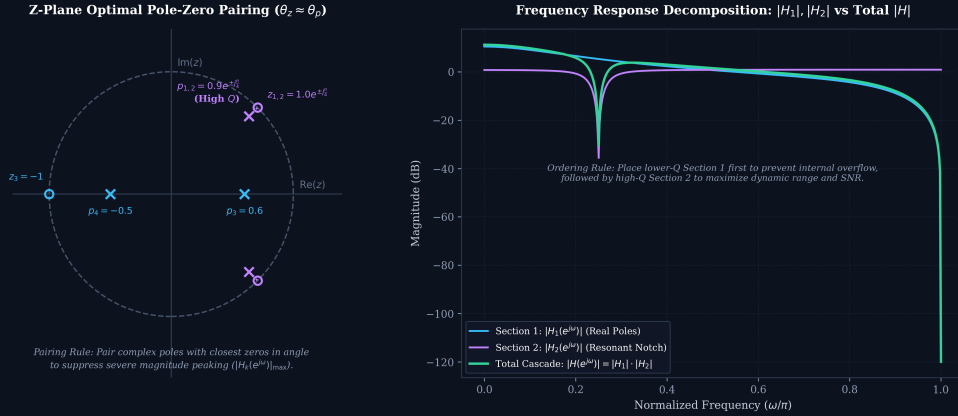


Figure 3: Optimal Pole-Zero Pairing Analysis. (Left) z -plane pole-zero geometry showing pairing of pole p_1 ($r = 0.9, \theta = 45^\circ$) with zero z_1 ($r = 1.0, \theta = 45^\circ$). (Right) Individual section magnitude responses showing how pairing suppresses extreme resonance peaks and maintains dynamic range balance.

4.2 4.2 Section Ordering (Sequencing)

Once the biquads $H_1(z), H_2(z), \dots, H_K(z)$ are formed, in what physical sequence should they be chained?

- **Standard Rule (Low-Q to High-Q):** Arrange sections in order of **increasing pole radius** ($r_1 \leq r_2 \leq \dots \leq r_K$).
 - Sections with poles further from the unit circle (broad bandwidth, low resonance) are placed first.
 - Highly resonant sections (poles closest to the unit circle, high Q) are placed last.
 - *Advantage:* Out-of-band noise and large input signal spikes are gradually attenuated by early sections before reaching the highly sensitive, resonant stages.
- **High-SNR Rule (Alternative):** Place the section with the highest gain first to boost low-level signals above the quantization noise floor, provided scaling prevents overflow.

4.3 4.3 Dynamic Range Gain Scaling (L_2 and L_∞ Scaling)

To prevent fixed-point arithmetic overflow at intermediate state nodes $w_k[n]$, a scaling factor g_k is distributed before each section:

$$g = \prod_{k=1}^K g_k \quad (18)$$

Using the L_∞ norm (peak frequency response) scaling:

$$g_k = \frac{1}{\max_{\omega} \left| \prod_{i=1}^k H_i(e^{j\omega}) \right|} \quad (19)$$

This guarantees that for any sinusoidal input with amplitude ≤ 1 , no intermediate node in the cascade will exceed unity magnitude.

5 5. HARDWARE & COMPUTATIONAL COMPLEXITY AUDIT

Table 1 provides a rigorous comparison between Direct Form realizations and Cascade Form realizations for an N -th order IIR filter (assuming $M = N$ even, $K = N/2$).

Architectural Metric	Direct Form I	Direct Form II	Cascade (DF-I SOS)	Cascade (DF-II SOS)
Canonical Status	Non-Canonical	Canonical	Non-Canonical	Canonical
Total Delay Registers (z^{-1})	$2N$	N	$2N$	N
Total Multipliers	$2N + 1$	$2N + 1$	$2N + 1$	$2N + 1$
Total Adders	$2N$	$2N$	$2N$	$2N$
Root Quantization Sensitivity	Catastrophic ($O(N!)$)	Catastrophic ($O(N!)$)	Extremely Low	Extremely Low
Stability Margin	Easily Lost	Easily Lost	Guaranteed Robust	Guaranteed Robust
Internal Overflow Control	None	Extremely Difficult	Moderate	Easily Scaled per SOS
Modularity for VLSI	Low (Irregular)	Low (Irregular)	High (Identical Cells)	Highest (Replication)

Table 1: Comprehensive Architectural Complexity and Performance Audit.

Crucial Takeaway for Examinations: The canonical Cascade Form requires the **exact same number of multipliers ($2N + 1$), adders ($2N$), and delay registers (N)** as Direct Form II, yet it completely eliminates the fatal root sensitivity and overflow vulnerability of high-order systems!

6 6. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

6.1 Example 18.1: Complete 4th-Order Cascade Realization with Mixed Roots

Problem Statement: An IIR digital filter is specified by the transfer function:

$$H(z) = \frac{(1 + z^{-1})(1 - \sqrt{2}z^{-1} + z^{-2})}{(1 - 0.9e^{j\pi/4}z^{-1})(1 - 0.9e^{-j\pi/4}z^{-1})(1 - 0.6z^{-1})(1 + 0.5z^{-1})} \quad (20)$$

- Determine all pole and zero locations and group them into conjugate and real pairs.
- Perform optimal pole-zero pairing based on frequency-domain proximity.
- Order the sections to minimize noise and overflow.
- Formulate the biquad transfer functions $H_1(z)$ and $H_2(z)$ with strictly real coefficients.

- (e) Write the complete difference equations for both stages using canonical Direct Form II.
- (f) Construct the overall hardware realization diagram.
- (g) Evaluate the first three output samples $y[0], y[1], y[2]$ for an impulse input $x[n] = \delta[n]$.

Solution:

(a) Pole-Zero Classification:

• **Poles:**

- Complex conjugate pair: $p_{1,2} = 0.9e^{\pm j\pi/4}$.

$$D_{\text{complex}}(z) = (1 - 0.9e^{j\pi/4}z^{-1})(1 - 0.9e^{-j\pi/4}z^{-1}) = 1 - 2(0.9)\cos(\pi/4)z^{-1} + 0.81z^{-2} \quad (21)$$

Since $\cos(\pi/4) = \sqrt{2}/2 \approx 0.7071$:

$$D_{\text{complex}}(z) = 1 - 1.2728z^{-1} + 0.81z^{-2} \quad (r = 0.9, \theta = 45^\circ, \text{High } Q) \quad (22)$$

- Real poles: $p_3 = 0.6, p_4 = -0.5$.

$$\begin{aligned} D_{\text{real}}(z) &= (1 - 0.6z^{-1})(1 + 0.5z^{-1}) \\ &= 1 - (0.6 - 0.5)z^{-1} - (0.6)(0.5)z^{-2} \\ &= 1 - 0.1z^{-1} - 0.3z^{-2} \end{aligned} \quad (23)$$

• **Zeros:**

- Complex zero pair on the unit circle:

$$N_{\text{complex}}(z) = 1 - \sqrt{2}z^{-1} + z^{-2} = 1 - 1.4142z^{-1} + z^{-2} \quad (24)$$

Roots: $z_{1,2} = e^{\pm j\pi/4}$ (radius $r_z = 1.0$, angle $\theta_z = 45^\circ$).

- Real zero: $z_3 = -1 \implies N_{\text{real}}(z) = 1 + z^{-1}$.

(b) Optimal Pole-Zero Pairing:

- The complex zero pair is at angle $\theta_z = 45^\circ$.
- The complex pole pair is also at angle $\theta_p = 45^\circ$ ($\Delta\theta = 0^\circ$).
- Pairing these two together provides exact frequency-alignment, allowing the transmission notch at 45° to suppress the resonant peak of the pole at 45° .
- The remaining real zero $(1+z^{-1})$ is paired with the real pole quadratic $(1-0.1z^{-1}-0.3z^{-2})$.

(c) Section Ordering: We order from lowest pole radius to highest pole radius (Low- Q to High- Q):

- **Section 1 (Biquad 1):** Real poles (maximum radius $r = 0.6$).
- **Section 2 (Biquad 2):** Complex poles (radius $r = 0.9$).

(d) Factored Transfer Functions:

$$H_1(z) = \frac{1 + z^{-1}}{1 - 0.1z^{-1} - 0.3z^{-2}} \quad (25)$$

$$H_2(z) = \frac{1 - 1.4142z^{-1} + z^{-2}}{1 - 1.2728z^{-1} + 0.81z^{-2}} \quad (26)$$

Overall transfer function: $H(z) = H_1(z) \cdot H_2(z)$.

(e) **Canonical Direct Form II Difference Equations:** Let $w_1[n]$ and $w_2[n]$ be the internal state variables for Section 1 and Section 2 respectively.

Section 1 Computations:

$$w_1[n] = x[n] - (-0.1)w_1[n-1] - (-0.3)w_1[n-2] \quad (27)$$

$$w_1[n] = x[n] + 0.1w_1[n-1] + 0.3w_1[n-2] \quad (\text{State Equation 1}) \quad (27)$$

$$y_1[n] = w_1[n] + w_1[n-1] \quad (\text{Output Equation 1}) \quad (28)$$

Intermediate Coupling Link:

$$x_2[n] = y_1[n] \quad (29)$$

Section 2 Computations:

$$w_2[n] = x_2[n] - (-1.2728)w_2[n-1] - (0.81)w_2[n-2] \quad (30)$$

$$w_2[n] = x_2[n] + 1.2728w_2[n-1] - 0.81w_2[n-2] \quad (\text{State Equation 2}) \quad (30)$$

$$y[n] = w_2[n] - 1.4142w_2[n-1] + w_2[n-2] \quad (\text{Output Equation 2}) \quad (31)$$

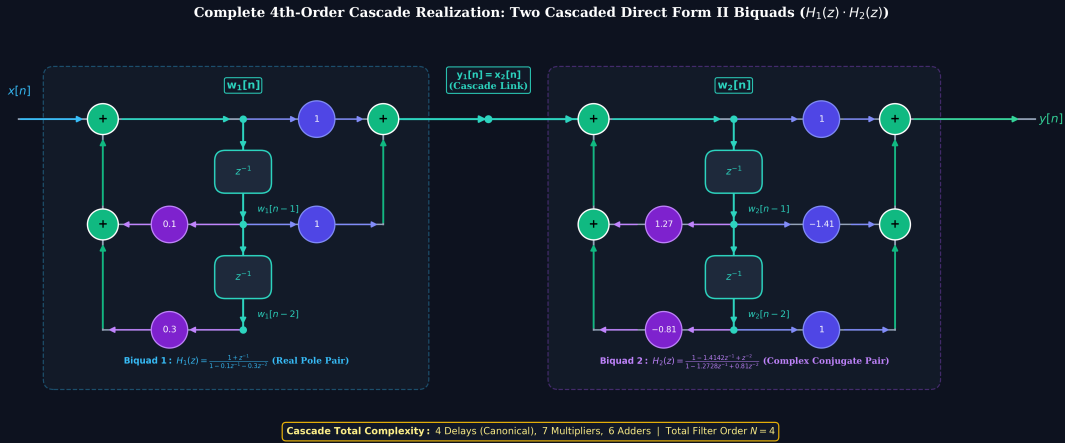


Figure 4: Complete 4th-Order IIR Cascade Realization (Example 18.1). Biquad 1 ($H_1(z)$) processes the real poles and zero, feeding intermediate sequence $y_1[n] = x_2[n]$ directly into Biquad 2 ($H_2(z)$), which realizes the complex conjugate poles and zeros.

(g) **Impulse Response Evaluation** ($x[n] = \delta[n]$): Assuming initially relaxed conditions ($w_1[-1] = w_1[-2] = w_2[-1] = w_2[-2] = 0$):

At $n = 0$ ($x[0] = 1$):

- $w_1[0] = 1 + 0.1(0) + 0.3(0) = \mathbf{1.0}$
- $y_1[0] = w_1[0] + w_1[-1] = 1.0 + 0 = \mathbf{1.0} \implies x_2[0] = 1.0$
- $w_2[0] = x_2[0] + 1.2728(0) - 0.81(0) = \mathbf{1.0}$
- $y[0] = w_2[0] - 1.4142(0) + 0 = \mathbf{1.0}$

At $n = 1$ ($x[1] = 0$):

- $w_1[1] = 0 + 0.1(1.0) + 0.3(0) = \mathbf{0.1}$

- $y_1[1] = w_1[1] + w_1[0] = 0.1 + 1.0 = \mathbf{1.1} \implies x_2[1] = 1.1$
- $w_2[1] = x_2[1] + 1.2728(1.0) - 0.81(0) = 1.1 + 1.2728 = \mathbf{2.3728}$
- $y[1] = w_2[1] - 1.4142w_2[0] + w_2[-1] = 2.3728 - 1.4142(1.0) + 0 = \mathbf{0.9586}$

At $n = 2$ ($x[2] = 0$):

- $w_1[2] = 0 + 0.1(0.1) + 0.3(1.0) = 0.01 + 0.30 = \mathbf{0.31}$
- $y_1[2] = w_1[2] + w_1[1] = 0.31 + 0.10 = \mathbf{0.41} \implies x_2[2] = 0.41$
- $w_2[2] = 0.41 + 1.2728(2.3728) - 0.81(1.0) = 0.41 + 3.0201 - 0.81 = \mathbf{2.6201}$
- $y[2] = w_2[2] - 1.4142(2.3728) + 1.0 = 2.6201 - 3.3556 + 1.0 = \mathbf{0.2645}$

The first three impulse response samples are $h[n] = \{\mathbf{1.0}, 0.9586, 0.2645, \dots\}$.

6.2 Example 18.2: Third-Order Odd System Realization (1st-Order + 2nd-Order)

Problem Statement: Realize the third-order IIR system in Cascade Form:

$$H(z) = \frac{0.5(1 - z^{-1})(1 + 0.8z^{-1} + 0.64z^{-2})}{(1 - 0.5z^{-1})(1 - 0.8z^{-1} + 0.5z^{-2})} \quad (32)$$

Solution: Because the system order is odd ($N = 3$), it must be decomposed into $K = \lceil 3/2 \rceil = 2$ sections:

- One First-Order Section ($H_1(z)$) realizing real pole $p_1 = 0.5$ and real zero $z_1 = 1$.
- One Second-Order Section ($H_2(z)$) realizing complex conjugate poles and zeros.

Section Definitions:

$$H_1(z) = 0.5 \left(\frac{1 - z^{-1}}{1 - 0.5z^{-1}} \right) \quad (33)$$

$$H_2(z) = \frac{1 + 0.8z^{-1} + 0.64z^{-2}}{1 - 0.8z^{-1} + 0.5z^{-2}} \quad (34)$$

Canonical Difference Equations: Stage 1 (First-Order Section):

$$w_1[n] = 0.5x[n] + 0.5w_1[n - 1] \quad (35)$$

$$y_1[n] = w_1[n] - w_1[n - 1] \implies x_2[n] = y_1[n] \quad (36)$$

(Requires 1 delay register, 2 multipliers, 2 adders).

Stage 2 (Second-Order Biquad):

$$w_2[n] = x_2[n] + 0.8w_2[n - 1] - 0.5w_2[n - 2] \quad (37)$$

$$y[n] = w_2[n] + 0.8w_2[n - 1] + 0.64w_2[n - 2] \quad (38)$$

(Requires 2 delay registers, 4 multipliers, 4 adders).

Total Hardware Count:

- Total Delays: $1 + 2 = \mathbf{3}$ registers (strictly canonical, $\max(M, N) = 3$).
- Total Multipliers: $2 + 4 = \mathbf{6}$ multipliers.
- Total Adders: $2 + 4 = \mathbf{6}$ adders.

6.3 Example 18.3: Reverse Synthesis — Extracting Transfer Function from Cascade SFG

Problem Statement: A cascaded filter has two biquads with intermediate equations:

$$\begin{aligned}\text{Biquad 1: } w_1[n] &= x[n] + 0.4w_1[n-1], & y_1[n] &= w_1[n] - 0.2w_1[n-1] \\ \text{Biquad 2: } w_2[n] &= y_1[n] - 0.6w_2[n-1], & y[n] &= 2w_2[n] + w_2[n-1]\end{aligned}$$

Find the overall system transfer function $H(z)$ and evaluate system stability.

Solution: Transforming to the Z -domain:

$$H_1(z) = \frac{Y_1(z)}{X(z)} = \frac{1 - 0.2z^{-1}}{1 - 0.4z^{-1}}, \quad H_2(z) = \frac{Y(z)}{Y_1(z)} = \frac{2 + z^{-1}}{1 + 0.6z^{-1}} \quad (39)$$

Overall transfer function:

$$H(z) = H_1(z) \cdot H_2(z) = \frac{(1 - 0.2z^{-1})(2 + z^{-1})}{(1 - 0.4z^{-1})(1 + 0.6z^{-1})} = \frac{2 + 0.6z^{-1} - 0.2z^{-2}}{1 + 0.2z^{-1} - 0.24z^{-2}} \quad (40)$$

Stability Check: The system poles are $p_1 = 0.4$ and $p_2 = -0.6$. Since $|p_1| = 0.4 < 1$ and $|p_2| = 0.6 < 1$, both poles lie strictly inside the unit circle, confirming that the system is **Bounded-Input Bounded-Output (BIBO) stable**.

7. FACULTY INSIGHTS, COMMON MISCONCEPTIONS & EXAM TRAPS

1. Misconception 1: "Cascade realization requires more delay elements than Direct Form II."

- **Reality: False.** A canonical cascade using Direct Form II biquads requires:

$$\sum_{k=1}^K \max(M_k, N_k) = N \quad \text{delay registers}$$

which is **identically equal** to the delay count of Direct Form II.

2. Exam Trap 1: The Minus Sign in Feedback Coefficients ($-a_{k1}, -a_{k2}$):

- In the denominator polynomial $1 + a_{k1}z^{-1} + a_{k2}z^{-2}$, if $a_{k1} = -1.2728$, students often mistakenly write -1.2728 in the state equation.
- **Correct Rule:** The state update is $w_k[n] = x_k[n] - a_{k1}w_k[n-1] - a_{k2}w_k[n-2]$. Therefore:

$$-a_{k1} = -(-1.2728) = +1.2728$$

Remember: In the signal flow diagram, the feedback multiplier is **negated** relative to the denominator polynomial coefficient!

3. Exam Trap 2: Does Section Cascading Order Matter?

- **Theoretical Answer:** In infinite-precision arithmetic,

$$H_1(z)H_2(z) = H_2(z)H_1(z)$$

because linear time-invariant systems commute under convolution.

- *Practical Hardware Answer: Yes, order matters critically!* In fixed-point digital signal processors with finite word-length, placing high- Q resonant sections first causes internal state overflow and massive quantization noise amplification. Always place low- Q sections first in physical hardware.

4. **Misconception 3: "Can a single complex pole be realized in a 1st-order section?"**

- *Reality: Never.* A 1st-order section $(1 - pz^{-1})$ with complex $p = re^{j\theta}$ produces complex coefficients, requiring complex arithmetic hardware ($4\times$ real multipliers). Complex poles **must always be paired with their complex conjugates** to form a second-order section with purely real coefficients.

8 8. SUMMARY & QUICK REVISION TABLE

Concept / Metric	Key Formula / Summary Rule
Cascade Transfer Function	$H(z) = g \prod_{k=1}^K H_k(z) = g \prod_{k=1}^K \frac{b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}}$
Total Sections (K)	$K = \lceil N/2 \rceil$ ($N/2$ biquads for even N ; $(N-1)/2$ biquads +1 first-order for odd N)
Biquad State Equation	$w_k[n] = x_k[n] - a_{k1}w_k[n-1] - a_{k2}w_k[n-2]$
Biquad Output Equation	$y_k[n] = b_{k0}w_k[n] + b_{k1}w_k[n-1] + b_{k2}w_k[n-2]$
Complex Pole Expansion	$1 - 2r_k \cos(\theta_k)z^{-1} + r_k^2 z^{-2} \implies a_{k1} = -2r_k \cos(\theta_k), a_{k2} = r_k^2$
Pole-Zero Pairing Rule	Pair closest pole-zero pairs in frequency ($\theta_z \approx \theta_p$) to suppress resonance peaks
Section Ordering Rule	Order sections by increasing pole radius r_k (Low- Q to High- Q) to maximize SNR
Canonical Hardware Count	N delay registers, $2N + 1$ multipliers, $2N$ adders (matches DF-II exactly)

Table 2: Lecture 18 Quick Revision Reference.